PHARMACOVIGILANCE · SYNTHETIC STUDY

Trusting signals in adverse-event reports

Anomaly detection, imputation, causal thinking and GenAI on safety-report data

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github.com/zahi-glitch/pv-signal-lab

OVERVIEW

The project in one page

Problem

Safety teams scan reports of side effects and flag drug–event pairs that look linked. Many flags are false, because of age, sex or missing data. Nobody knows which flags are real.

What I did

Simulated reports where the truth is known. Built the standard detection rules, an age adjustment, imputation, a sex analysis and a label assistant. Measured how well each one works.

What I found

Age makes 83% of non-causal pairs look like signals; adjusting cuts that to 1%. Poor imputation brings the problem back. Pooling hides signals that act in one sex. The assistant refused every unanswerable question.

All data are synthetic · each claim is tested in the repository

What is a safety signal?

Report

One patient: age, sex, the drugs taken and the reactions reported. Voluntary and unverified.

Signal

A drug and an event reported together more often than expected. A hypothesis, not proof.

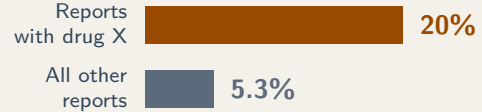
Why it is hard

No count of exposed patients, many co-medications, and age or illness can create associations on their own.

Safety teams screen thousands of drug–event pairs and review the flagged ones. The question is which flags to trust.

Example: drug X and rash

1. A database holds 2,000 reports.
2. 100 mention drug X. Of these, 20 report a rash: 20%.
3. Of the other 1,900 reports, 100 report a rash: 5.3%.



Share of reports that mention a rash

Rash is reported about four times as often with drug X. That makes the pair a candidate signal, to be reviewed by people.

Reviewers then check timing, dose, other drugs taken, and age or illness before acting. The numbers are invented and reused on the next slide.

How a signal is computed

	rash	no rash
drug X	**20** a	80 b
other drugs	100 c	1,800 d

Counts of reports; a, b, c, d are the four cells. a: drug and rash. b: drug, no rash. c: other drugs, rash. d: other drugs, no rash.

Rash rate with drug X: $20/100 = 20\%$

Rash rate without it: $100/1{,}900 = 5.3\%$

PRR $= 20\%/5.3\% = \mathbf{3.8}$

How many times more often the event is reported with the drug.

ROR $= (20 \times 1{,}800)/(80 \times 100) = \mathbf{4.5}$

The same comparison, made with odds instead of rates.

Is that more than chance? Test $\ln(\mathrm{ROR})$, and control false discoveries across all tested drug–event pairs at once (1,200 in the simulation: 30 drugs $\times$ 40 events).

Worked example from the unit test in tests/test_signals.py · definitions in the glossary

Why a flagged pair may not be real

Confounding

A third factor, such as age, makes drug and event appear together with no causal link.

Effect modification

The drug acts in one group only, such as one sex. A pooled number then describes nobody.

Missing data

Fields such as age are often absent. The way they are filled in can bring false signals back.

In real databases nobody knows which flags are true, so nobody can measure how well these safeguards work. The next slide shows a way around that.

Build the truth, then measure

Simulated reports have a known answer: which pairs are true signals and which only look like it. So for every method we can count how many true signals it finds and how many false ones it raises.

Q1
Age
Does age create false signals, and does adjusting for it fix that?

Q2
Missing ages
What happens to the adjustment when some ages are missing?

Q3
Sex
What if a drug acts in one sex only?

Q4
GenAI
Can a label assistant refuse when its sources do not cover the question?

3 METHODS

Simulate, count, detect, score

1
Simulate

Reports with age, sex, drugs and events; 8 injected signals and 36 age-confounded pairs.

2
Count

A 2×2 table for every drug–event pair, in Python and in plain SQL.

3
Detect

Evans rule, FDR-controlled ROR test, the same test stratified by age, and per sex.

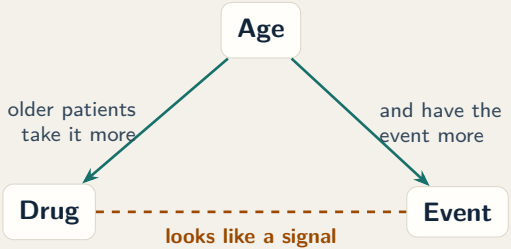
4
Score

Flags compared with the known truth: signals found, false alarms.

Every result comes from repeating this on many simulated databases.

Evans rule: PRR ≥ 2, ≥ 3 co-reports, $\chi^2 \geq 4$ · FDR: Benjamini–Hochberg, q $= 0.05$

How age fakes a signal, and the fix



Older patients take some drugs more and also have some events more often. The two then co-occur with no causal link.

Fix: compare like with like. Split the reports into five age bands, compute the odds ratio inside each band, and combine them (Mantel–Haenszel).

4 RESULTS · Q1 AGE

Age creates false signals

Question. Does age create false signals, and does adjusting for it fix that?

83%
of confounded pairs flagged by the crude test

1%
flagged after stratifying by age band

0.98
share of true signals found, by both tests

Same databases and test; only the age adjustment differs.

About 24,000 reports per database · mean over 20 databases · run_detection.py

Adjusting costs little sensitivity

Age adjustment costs up to seven points of sensitivity in small databases and removes nearly all confounded false flags.

Reports kept	Crude: found	Crude: false	Adjusted: found	Adjusted: false
2,959	0.681	**0.215**	0.612	0.001
5,917	0.794	**0.406**	0.750	0.000
12,307	0.906	**0.654**	0.875	0.004
24,130	0.981	**0.826**	0.981	0.010

found = true signals flagged · false = confounded pairs flagged · mean over 20 databases per size

Missing ages come in three kinds

Question. What happens to the age adjustment when some ages are missing, and what is the best way to fill them in?

MCAR

completely at random

A field is lost by chance. Nothing predicts it.

MAR

at random, given recorded data

Depends on what we do record. Here: sex and number of drugs.

MNAR

not at random

Depends on the missing value itself. Here: older patients are recorded less often.

Age is the confounder, so a poor fix puts the false signals back. Compared next: drop the reports, fill with the mean, copy a random age (hot-deck), or predict age (regression).

Simple imputation brings bias back

Strategy	MCAR	MAR	MNAR
Ignore age	**0.84**	**0.84**	**0.84**
Complete cases	0.01	0.01	0.00
Mean	**0.28**	**0.38**	**0.49**
Hot-deck	**0.44**	**0.49**	**0.69**
Regression	0.01	0.02	0.03
True ages	0.01	0.01	0.01

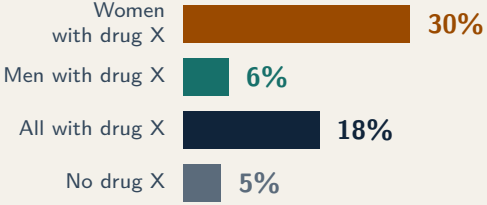
Share of confounded pairs flagged, lower is better. Mean and hot-deck imputation bring the confounding back, most under MNAR.

Mean imputation has the lowest age error (18.3 vs 20.5 years for regression, MCAR), yet it does not protect the test. Regression works because drugs and events are in its model.

30% of ages removed · 10 databases · experiments/run_imputation.py

Same drug, different effect by sex

Question. What if a drug raises the event risk in one sex only?



Ratio of rates: women $30/5 = \mathbf{6.0}$, men $6/5 = \mathbf{1.2}$, everyone together $18/5 = \mathbf{3.6}$.

The pooled number describes nobody. This is **effect modification**. Unlike age confounding, there is nothing to adjust away.

Method: report the signal **per sex** and **test whether the sexes differ**. Numbers invented for illustration, equal numbers of women and men.

Pooling hides one-sex signals

Reports	Found: pooled	Found: by sex	Sex named	Difference detected
20,000	**62%**	70%	70%	91%
40,000	**80%**	86%	86%	88%
80,000	91%	98%	98%	99%

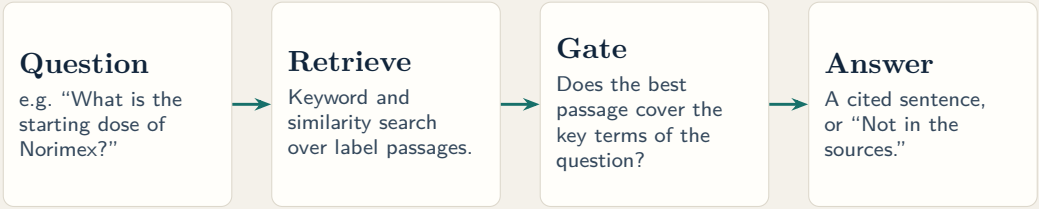
Four of eight true signals act in one sex only. Pooling dilutes them; the by-sex analysis finds more and names the right sex whenever it flags one.

Cost: twice as many tests, so null false alarms double, to about 0.1%.

Difference = Wald test on ln ROR, $p < 0.05$; fires on 4–6% of both-sex signals · 20 databases per size, no age confounding, few true signals, so noisy · run_sex.py

How the label assistant works

Question. Can an assistant answer from drug labels and refuse when the labels do not cover the question?



A model that always answers will sometimes invent. Refusing is a design decision, and it can be measured. A language model can be plugged in after the gate, and is never called when the gate refuses.

It refuses what it cannot source

Tested on 14 fictional label passages and 24 questions, 10 of which the sources cannot answer.

0%
of 10 unanswerable questions answered

0
wrong answers across all 24 questions

64%
of 14 answerable questions answered correctly; 36% refused

A demonstration of the evaluation method, not a benchmark · experiments/run_rag_eval.py

5 TAKEAWAYS

What the project shows

Q1 Age alone makes the crude test flag 83% of non-causal pairs. Stratifying by age band cuts that to 1%, at a small cost in sensitivity.

Q2 Missing ages need care: complete cases and regression imputation stay at 0–3%, mean and hot-deck bring back 28–69%.

Q3 Pooling hides signals that act in one sex. Per-sex analysis finds more of them and a heterogeneity test flags the difference.

Q4 A label assistant can be built to refuse: 0 of 10 unanswerable questions answered, no wrong answers, at the price of refusing 36% of answerable ones.

Before trusting a flag, ask what could produce it without a causal link, and measure that.

All results from synthetic data

SKILLS MAP

Requirement and evidence

Requirement	Evidence in the repository	Status
Anomaly detection	Signal detection with false-discovery control	**Done**
Data imputation	Missing ages: mean, hot-deck, regression	**Done**
Projection	Not covered yet	**Gap**
Statistical and ML methods	PRR/ROR, Mantel–Haenszel, ridge imputation	**Done**
Causal analysis	Age confounding; sex as effect modifier	**Partly**
GenAI applications	Label assistant: citations, abstention	**Done, toy data**
Python and SQL	Tested package; counts in SQL (SQLite)	**Done**
Azure	Deployment on the roadmap	**Planned**
Pharma data	Synthetic reports; openFDA loader untested	**Partly**

Requirements follow the Data Scientist, GenAI & Advanced Analytics posting

LIMITS

What this study cannot show

Synthetic data. Real reports add duplicates, free-text drug names and more confounders than age.

Reports are not incidence. A flag is a hypothesis for clinical review, not evidence of causation.

Two factors only. The simulation has age (confounder) and sex (effect modifier). Co-medication is missing.

Real-data loader untested. The openFDA loader was not run against the live API.

Defaults, not tuned. The Evans rule and a false-discovery rate of 0.05 are used as published.

All numbers in this deck come from the synthetic experiments in the repository

NEXT

Next steps

1. Run the pipeline on real openFDA reports for one drug class and review the top flags against the labels
2. Add empirical-Bayes shrinkage to the method comparison
3. Add co-medication as a confounder, with a propensity-style adjustment
4. Add a projection step, for example forecasting report volumes
5. Replace the keyword retriever with dense embeddings and test on a larger question set from real labels
6. Deploy the assistant as a small cloud service on Azure

github.com/zahi-glitch/pv-signal-lab

APPENDIX

Glossary

Signal	A drug–event pair reported more often than expected
PRR	Event rate with the drug divided by the event rate without it
ROR	Odds of the event with the drug divided by the odds without it
FDR	Controls the expected share of false flags among all flags (here 5%)
Confounder	A third factor, here age, that drives both drug use and the event
Mantel–Haenszel	An odds ratio combined across strata such as age bands
Effect modifier	A factor, here sex, that changes the size of a drug's effect; report it per group
MCAR, MAR, MNAR	Missing by chance, given recorded data, or depending on the value itself
Imputation	Filling a missing value with an estimate
RAG	Retrieval-augmented generation: answer from retrieved sources
Abstention	Refusing to answer when the sources do not cover the question